Prove that $(2n^2 + 3n + 4) = O(n^4)$ $2n^2 + 3n + 4 \leq 2n^4 + 3n^4 + 4n^4$ when $n \geq 1$ $2n^2 + 3n + 4 \leq 9n^4$ So, $c = 9$ and $n_0 = 1$ $\therefore (2n^2 + 3n + 4) = O(n^4)$	Right-side verification: $2n^2 \leq 2n^4$ $1 \leq n^2$ $n \geq 1$ $3n \leq 3n^4$ $1 \leq n^3$ $n \geq 1$ $4 \leq 4n^4$ $1 \leq n^4$ $n \geq 1$

Prove that $\log_4(4n^2 - 5n + 8) = O(\log_4(n))$.	
$\log_4(4n^2 - 5n + 8) \le \log_4(4n^2 + 8)$	(when bounding above, drop negative terms)
$\le \log_4(4n^2 + 8n^2)$	$8 \le 8n^2$ $1 \le n$
$\le \log_4(12n^2)$	
$\le \log_4(n^3)$	$12n^2 \le n^3$ when $n \ge 12$
$\le 3\log_4(n)$	
So, $C = 3$ and $n_0 = 12$	
$\therefore \log_4(4n^2 - 5n + 8) = O(\log_4(n))$	

Prove that $\log_4(4n^2 + 5n + 8) = O(\log_4(n))$.	
$\log_4(4n^2 + 5n + 8) \le \log_4(4n^2 + 5n^2 + 8n^2)$ $\le \log_4(17n^2)$ $\le \log_4(17) + \log_4(n^2)$ $\le \log_4(17) + 2\log_4(n)$ $\le \log_4(n) + 2\log_4(n)$ $\le 3\log_4(n)$	$5n \le 5n^2$ $(n \ge 1)$ $8 \le 8n^2$ $(n \ge 1)$
Here, $C = 3$ and $n_0 = 17$	when $n \ge 17$
$\therefore \log_4(4n^2 + 5n + 8) = O(\log_4(n))$	

Prove that $(2n^2 + 3n + 4) = \Omega(n^2)$. $2n^2 + 3n + 4 \geq 2n^2$ So, $C = 2$ and $n_0 = 0$ $\therefore (2n^2 + 3n + 4) = \Omega(n^2)$	Drop positive lower order terms

Prove that $(2n^2 + 3n + 4) = \Omega(n)$. $2n^2 + 3n + 4 \geq 3n$ Here, $C = 3$ and $n_0 = 0$ $\therefore (2n^2 + 3n + 4) = \Omega(n)$	

Prove that $\log_4(4n^2 + 5n + 8) = O(\log_4(n))$.	
$\log_4(4n^2 + 5n + 8) \leq \log_4(4n^2 + 5n^2 + 8n^2)$	$5n \leq 5n^2$ $(n \geq 1)$
$\leq \log_4(17n^2)$	$8 \leq 8n^2$ $(n \geq 1)$
$\leq \log_4(17) + \log_4(n^2)$	
$\leq \log_4(17) + 2\log_4(n)$	
$\leq \log_4(n) + 2\log_4(n)$	
$\leq 3\log_4(n)$	
Here, $C = 3$ and $n_0 = 17$	when $n \geq 17$
$\therefore \log_4(4n^2 + 5n + 8) = O(\log_4(n))$	

Prove that, $\sqrt{(6n^3 + 7n^2 + 3n + 5)} \in \Theta(n^{1.5})$ **Upper Bounding :** $\sqrt{(6n^3 + 7n^2 + 3n + 5)} \leq \sqrt{(6n^3 + 7n^3 + 3n^3 + 5n^3)}$ $\leq \sqrt{(21n^3)}$ $\leq \sqrt{21} \cdot n^{1.5}$ Here, $C_2 = \sqrt{21}$ and $n_0 = 1$ So, $\sqrt{(6n^3 + 7n^2 + 3n + 5)} = O(n^{1.5})$ **Lower Bounding:** $\sqrt{(6n^3 + 7n^2 + 3n + 5)} \geq \sqrt{(6n^3)}$ $\geq \sqrt{6} \cdot n^{1.5}$ Here, $C_1 = \sqrt{6}$ and $n_0 = 0$ So, $\sqrt{(6n^3 + 7n^2 + 3n + 5)} = \Omega(n^{1.5})$ $\therefore \sqrt{(6n^3 + 7n^2 + 3n + 5)} = \Theta(n^{1.5})$	$7n^2 \leq 7n^3$ $(n \geq 1)$ $3n \leq 3n^3$ $(n \geq 1)$ $5 \leq 5n^3$ $(n \geq 1)$